



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## APACHE SPARK PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

**Q1** What is Apache Spark and what problem in Data Engineering does it solve?

Threat Model

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**Q6** How does Apache Spark handle reliability and high availability?

Observability

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**Q2** What are the core components or concepts in Apache Spark?

Architecture

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**Q7** What observability does Apache Spark provide out of the box?

Lifecycle

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**Q3** How does Apache Spark compare to its main alternatives?

Configuration

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**Q8** What are common performance bottlenecks in Apache Spark?

Compliance

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**Q4** How do you get started with Apache Spark for a new project?

Hardening

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**Q9** What security best practices apply when running Apache Spark?

Testing

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**Q5** What configuration options most affect Apache Spark's behavior in production?

SSO / IdP

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**Q10** What mistakes do teams commonly make when adopting Apache Spark?

Tuning

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## ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

### A1 Apache Spark is a tool or

Mitigation

Apache Spark is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

### A6 Apache Spark provides HA through leader

Observability

Apache Spark provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

### A2 Apache Spark has key building blocks

Primitives

Apache Spark has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

### A7 Apache Spark exposes metrics (Prometheus)

Automation

Apache Spark exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

### A3 Apache Spark trades off simplicity against

Hardening

Apache Spark trades off simplicity against feature richness differently than its alternatives. Choose Apache Spark when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

### A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

### A4 Install Apache Spark via its package

Gotchas

Install Apache Spark via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

### A9 Run Apache Spark with the principle

Assessment

Run Apache Spark with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

### A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

### A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.